

Proposal of a fuzzy logic controller for the improvement of irrigation scheduling decision-making in greenhouse horticulture

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Abstract—A fuzzy logic controller is proposed for the evaluation of water drainage in order to determine if water content is easily available to crops; the lexical uncertainty given by “easily available” was quantified into a numerical value in this first approach. The result is a preliminary controller that embeds structural human knowledge about irrigation scheduling into an analytic fuzzy logic model (FLM), in order to provide agricultural scientists with quick access to a particular crop’s main production and growth related variables, and allowing for future data driven decisions on the spot. The controller was validated with real data collected from a particular crop, using remote sensors connected to Internet.

Index Terms—Decision-making system, agriculture in protected environments, fuzzy logic model, greenhouse horticulture, irrigation scheduling

I. INTRODUCTION

Recent studies have shown that climate change is affecting water availability everywhere [1], which has a direct impact on agricultural activities. This is the case in greenhouse horticulture soil-less (substrate) crops, where optimal irrigation scheduling is needed to positively influences the plants’ root zone environment and their photosynthesis process. There are two types of irrigation methods: conventional and proportional. Conventional irrigation is applied during long periods and with high quantities of water, with a resulting negative impact in soil or substrate salinity. Proportional irrigation uses shorts periods of time and is applied frequently; this procedure reduce waste of water and losses from pests and diseases, improving crop production. The knowledge of consumptive use of major crops, known as evapotranspiration (ET), is a prerequisite for efficient irrigation because the amount of ET can be related to the water supplied by the irrigation system, the change in water stored in substrate and the amount of water drained out of the plant. The water storage capacity depends strongly in growing medium, soil-less medium has less water storage capacity than soil. Water and nutrients are wasted when applied during a period of time when relative humidity in the greenhouse remains at 100%; for example, when the air temperature and dew point temperature are similar [9]. Greenhouse horticulture cultivation reduces ET to about 70% at open field, thus improving the water usage relative to unprotected environment cropping [2]. In general, an irrigation

system should be controlled in order to provide enough water to plants, taking transpiration close to its potential value. Substrate moisture content measurements are evaluated in order to make decisions about irrigation: a major emphasis is given in the specialized literature to the term: “water availability”, which limits plant transpiration, meaning that the irrigation scheduling must replace the water lost by the plants’ water uptake and evaporation from the root zone. On the other hand, a deficiency in water supply may improve the level of crop stress. Researchers and producers need to ensure that the crop stress, observed or measured, is due to a deficit of moisture at the root of the plant, and not because of a disease, a pest or exceptional atmospheric conditions [3].

One of the major problems faced by agricultural scientists and producers, when dealing with decisions aimed at improved crops yield or water usage, is the blurred boundaries between such lexical criteria as established in the technical literature. Besides, many of the percentages used to determine water or fertilizer quantities are based on rules of thumb, which sometimes are derived from years of experience on particular crops on particular environments. This makes difficult not only to compare or test the performance of any proposed method of irrigation or fertilization in heterogeneous environments, but also make cumbersome and error prone the tasks of monitoring and decision-making. And while the reliable measurement of variables related to crop growth and yield is not new nor expensive anymore thanks to the Internet (see [4], [5]) not many abstract models are still available that may serve as quick decision tools for users.

Now, non-statistical uncertainty, or fuzziness, is known as an inherent property of a system that most of the time cannot be resolved by simple observations. Fuzzy Logic Theory (FLT) generally establishes a mathematical frame which will model human cognitive processes in a way that can be solved by a computer. Some authors as Zadeh define Fuzzy Logic (FL) as a logical system, which is an extension of multivalued logic that is intended to serve as a logic for “approximate reasoning”. The two most important concepts within FL are that of a linguistic variable and the fuzzy *if-then* rule. Linguistic variables allow the translation of natural language into logical, or numerical statements, which provide the tools for approximate reasoning. This means that, if measurement data is available,

and a fuzzy model is correctly defined, incorporating the human language rules of the plant's water availability and its implications regarding irrigation, for a particular crop is straightforward.

This work proposes a new way of including drainage collection data in order to propose an irrigation fuzzy decision model that provides controlling scalar outputs, which may later serve as the basis of an expert system.

This paper is organized as follows: Section II describes the problems regarding the impact of climate change and water in relation with greenhouse horticulture. Section III presents the methodology and implementation of the fuzzy logic variables and rules, and the resulting fuzzy Mamdani controller, with particular emphasis given to drainage measurements and irrigation scheduling decision-making. In Section IV some results are given, related to a particular example using the QtFuzzyLite software, with a proposal on a more extensive validation of the controller. Finally, some conclusions and perspectives are given in Section V.

II. PROBLEM DESCRIPTION

A. Drainage relationship with irrigation

Typically, the main environmental factors affecting plant growth are: light, temperature, water humidity, relative humidity and nutrition. Some authors have designed fuzzy controllers considering environmental variables for smart greenhouses [7], [8], but none of them have evaluated a drainage measurement as a decision-making for irrigation.

The drainage of a crop has two basic functions: not allowing salt accumulation in the substrate, taking away excessive accumulated nutrients; and reestablishing the nutrients' concentration in the substrate. The drainage can also be used to determine substrate humidity (SH) in hydroponic cultivation. Excess drainage implies losses of water and fertilizers that have a direct impact on the environment. With low levels of drainage, in a tropical ecosystem as Costa Rica's, there exists the risk of substrate salinization, and taking in consideration that plants absorb water at a major velocity than nutrients, this may cause excessive salt accumulation in the substrate. Some authors [9] have established a practical criteria about irrigation taking into consideration the "drainage percentage"(DP) and using it for decision-making about deficient or excessive irrigation. This DP is simply calculated as

$$DP = \left(\frac{\text{Sol. Drained}(L/m^2)}{\text{Sol. Applied}(L/m^2)} \right) \quad (1)$$

where *Sol. Drained* corresponds to the quantity of the drained solution collected and *Sol. Applied* the quantity of applied water with fertilizers or nutrients added. The value of the DP generally depends on the quality of water used for irrigation. If the water salt content increases so does the DP, up to 30-40%. Inversely, if water quality is good, producers have the possibility to reduce the DP up to 10 or 15%, which translates in lower production costs, because less water and nutrients need be applied. Another advantage of drainage collecting is

that this water can be reused as irrigation, saving also in the nutritive solutions applied [9].

In a typical horticulture production, overdrain, pH and conductivity measurements, are collected by hand. This process may be improved by using remote sensing that sends data to the "cloud" (see Fig. 1), but this only means that, instead of tabulating data by hand, researchers and producers end up with tons of unprocessed data in their computers, that still requires analysis in order to yield useful information.

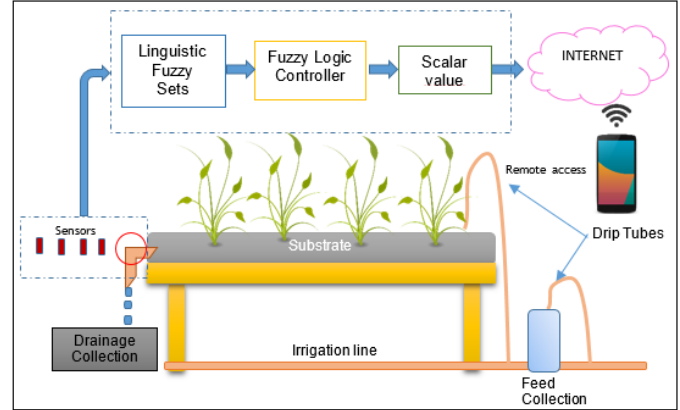


Figure 1. Schematic representation of a typical greenhouse horticulture vegetable production. Usual points of drainage collection and irrigation are shown. The place where the proposed fuzzy controller is to be located is also shown.

III. METHODOLOGY AND IMPLEMENTATION

An affordable data collection unit has been shown in [4], [5] where a hardware and software description is made. This system allows for the constant measurement of environmental variables (such as ambient temperature, humidity and light intensity) as well as substrate moisture, drainage, pH and conductivity. The data is then loaded to the cloud for instant monitoring and processing. The system's performance has been tested both in a controlled greenhouse, and in a microalgae culture, where acquired data from environmental variables, substrate moisture conditions and overdrain collection, are sent over the Internet to be processed using data fusion techniques and expert based criteria.

Therefore, the means of having information for decision-making on, for instance, how to minimize irrigation process frequency, required quantity of water, nutrients and fertilizers during the whole span of growth and production of a crop is available; this, of course, if data is handled with models that are capable of fusing it with human expertise. Here is where the use of FLM to translate lexical common sense rules into numerical values, considering the crop drainage data, may provide researchers and producers with a useful tool for efficiently planning their irrigation scheduling. The next steps were: first, to define which linguistic variables were related to the irrigation process; and second, to establish the necessary fuzzy sets with their related fuzzy rules.

A. Selection of the fuzzy inference systems

A fuzzy inference system (FIS) uses fuzzy set theory to map inputs features to outputs classes; both in the case of fuzzy classification. There are two types of FIS: Mamdani and Sugeno. A Mamdani FIS finds the centroid of a two dimensional model while a Sugeno's may be used to model any inference system in which the output membership functions are either linear or constant. In this case, a Mamdani FIS controller was selected due to the binary nature of the process under consideration (see [6], [8] for other examples of Mamdani controllers).

Now, since a fuzzy set (FS) is a set containing elements that have varying degrees of membership, it allows for the description of concepts with a certain ambiguity. A FS admits the possibility of partial membership in it and the degree an object belongs to a FS is denoted by a membership value between 0 and 1. In short, inputs, outputs and rules definition are to be established in order to model a complete FL Inference System. In Table I there is a description of all the names and ranges considered in the model proposed. Since decisions are binary, a triangular function has been used here for each fuzzy set, with the triangles' centroids serving as the decision-making point regarding drainage or irrigation.

B. Hedges and fuzzy rules

A hedge serves as a modifier of fuzzy values. There are five different kinds of hedges: concentration (very), dilation (somewhat), contrast intensification (extremely), vague (seldom) and probabilistic (likely). In the presented model the concentration hedge *very* was used to establish a difference in membership regarding water availability. The concentration hedge is here defined, in general terms, as

$$\mu_{A'}(x) = \mu_A(x)^p, p > 1 \quad (2)$$

where A' is the concentration of set A .

The hedge *very* can be implemented using a square function, that is, for *Water Very Difficult Available*- WVDA and *Water Difficult Available* - WDA, this may be represented as

$$\mu_{WVDA}(x) = \mu_{WDA}(x)^2 \quad (3)$$

Now, in real life, there are rules about reasoning and possible consequences. The antecedents of a fuzzy rule are propositions containing linguistic variables. In general, a fuzzy rules is expressed as in

$$\text{if } A \text{ is } a \text{ and } B \text{ is } b \text{ then } C \text{ is } c \quad (4)$$

where A and B are fuzzy sets with a universe of discourse X_1 , and C is a fuzzy set with a universe of discourse X_2 . Fuzzy reasoning components are separated into fuzzification (definition of inputs, outputs and associated fuzzy rules), inferencing (series of fuzzified output values) and defuzzification (converting the output of the fuzzy rules into a scalar or non fuzzy value). The *if-then* rules describe situations that can happen, with a corresponding action that should be executed. It is possible that more than one situation, as described by *if-then* rules, are simultaneous active, with different actions.

Table I
FUZZY SETS AND FUZZY RULES DEFINITION FOR THE PROBLEM OF IRRIGATION DECISION-MAKING BASED ON THE LINGUISTIC CONCEPT OF WATER AVAILABILITY.

Drainage- Input Name and range	Irrigation - Output Name and range	Fuzzy Rules
Water Very Difficult Available (WVDA) 0.000 0.200 0.400	NoIrrigation 0.000 0.200 0.400	If Drainage is WVDA then Irrigation is HighIrrigation
Water Difficult Available (WDA) 0.150 0.350 0.550	LowIrrigation 0.200 0.400 0.600	If Drainage is WDA then Irrigation is MediumIrrigation
Water Medium Difficult Available (WMDA) 0.300 0.500 0.700	MediumIrrigation 0.400 0.600 0.800	If Drainage is WMDA then Irrigation is MediumIrrigation
Water Easily Available (WEA) 0.450 0.650 0.850	HighIrrigation 0.600 0.800 1.000	If Drainage is WEA then Irrigation is LowIrrigation
Water Very Easily Available (WVEA) 0.600 0.800 1.000		If Drainage is WVEA then Irrigation is NoIrrigation

C. Implementation of the fuzzy logic controller

In this work, all the methodological steps for defining a Mamdani-type controller were followed. Table I shows the definition of the membership functions for each input variable (in this case "water drainage"), with the defined fuzzy rules and the corresponding outputs. The resulting controller provides now with a scalar value that maps to a numerical value the linguistic variables related to irrigation scheduling decision-making.

IV. RESULTS AND DISCUSSION

A validation of the preliminary model was carried out with data from a three month period of observation of a cucumber crop, from seeding all the way to the final harvest. This data not only included the measured input variables of the system (drainage, temperature, conductivity, pH) but also had recorded the decisions taken by the researchers regarding irrigation and fertilization, using the same measured variables. This provided with an initial point of comparison and adjustment for the model.

The QtFuzzyLite software tool was used to build and test the FIS. Figure 2 shows five triangular functions representing input fuzzy sets for Water Availability Membership (as defined in Table I). As an example, drainage input function has a value of 0.705 for WMDA and 0.545 for WEA, as is expressed by

$$\mu_{Drainage}(x) = 0.705_{WMDA} + 0.545_{WEA} \quad (5)$$

To determine the best action to take about irrigation process, the max-min inference method was selected and the rule with the largest firing strength was determined to infer an action from a set of activated situations. The centroid of the area under that function is calculated and the horizontal coordinate of that centroid is taken as the output of the Mamdani controller.

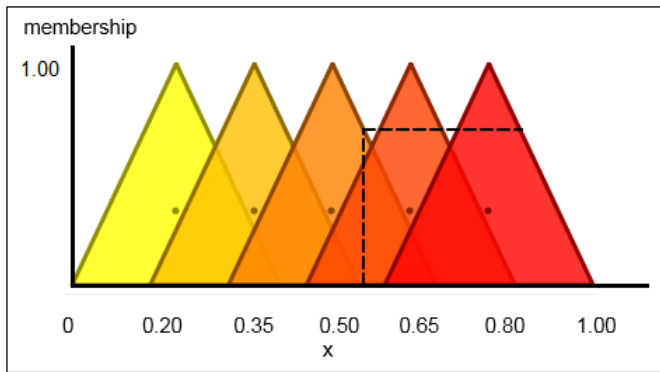


Figure 2. Water Availability Membership function. Five linguistic variables are defined, with a triangular function defined for each fuzzy set (image created with QtFuzzyLite).

For our example, the largest firing strength is 0.705, which corresponds to the MediumIrrigation membership function. Figure 3 illustrates the calculation of the output, using

$$\mu_{Irrigation}(y) = 0.545_{LowIrrig} + 0.705_{MediumIrrig} \quad (6)$$

where all zero membership fuzzy sets are not represented.

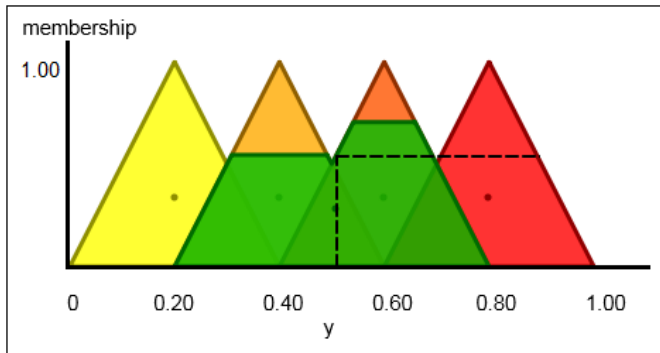


Figure 3. Example of the Irrigation Membership function. Calculated output value corresponding to inputs given in Figure 2 are shown, where a triangular function is defined for each fuzzy set (image created with QtFuzzyLite).

The results provided by the controller were then compared, for the experimental input data gathered, with the actions carried out by the researchers during crop production. A major divergence was found in several occasions which pointed to a typical problem when translating human based reasoning: when confronted on why researchers had decided not to apply high irrigation when predicted by the controller in several cases when water was in the order of "Medium difficult available", they pointed to the particularly humid and warm conditions of the environment on such occasions. Several other cases were found when the pH levels indicated excessive saltiness, which prompted a more aggressive water irrigation when the model proposed otherwise. Such results were somewhat expected, and are being used to expand the inference rules for a better tuned controller. To achieve better results, nonetheless, a parallel evaluation of the controller with an

ongoing crop production is mandatory. This should not only help in tuning the inference system by introducing other expert criteria as a particular situation arises, but would also give the chance to set up a control group of plants grown under the same conditions, except for the irrigation control. With this, real results regarding plant growth and produce yield should provide with a clearer picture about the usefulness of the proposed fuzzy controller.

V. CONCLUSIONS

An example of a fuzzy inference system for drainage and irrigation control has been shown; concentration hedges "very" were considered in order to establish fuzzy rules to determine an approximate scalar value to represent the action to be taken (in this case to know how much irrigation should be applied). A Mamdani fuzzy controller with a maximum method and centroid output was selected as a first approach. For a firing strength of 0.705, for instance, given by the maximum inference method and using the proposed Mamdani centroid controller, the action should be 0.705 of the Medium Irrigation set point.

In the future, more efficient fuzzy models of irrigation scheduling decision-making may be implemented by considering variables such as: substrate moisture, drainage pH and conductivity, and environmental temperature and humidity; all variables that add up to the inference rules required to anticipate the water needs of plants during their growing process, therefore maximizing their production while minimizing water, nutrients and fertilizer usage. For this, the evaluation of the controller with an ongoing crop production is mandatory, where a control group of plants is also established to provide with a more rigorous testing of the proposed controller.

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